Visualize Multiple Access Techniques



Differences between various multiple access techniques.

Frequency Division Multiple Access (FDMA):

- **Concept:**

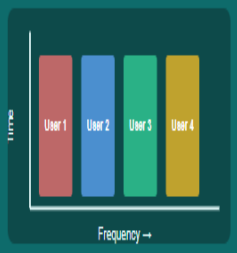
  Allocates different frequency bands to individual users.

  Each user is assigned a unique frequency band for communication.

- **Representation:**

  In a visual representation, you can use rectangles for represent different frequency bands.

  Label each rectangle with the corresponding user or communication channel.



- **Key Points:**

  Users transmit continuously on their allocated frequency band.

  Suitable for analog signals and FDMA is commonly used in analog radio communication.

Time Division Multiple Access (TDMA):

- **Concept:**

  Divides the time axis into slots or frames.

  Each user is assigned a specific time slot during which it can transmit.

- **Representation:**

  Visualize time slots as distinct sections within a timeline.

  Label each section with the corresponding user or device.



- **Key Points:**

  Users take turns to transmit during their designated time slots.

  Efficient for digital communication, commonly used in GSM (Global System for Mobile Communications).

Code Division Multiple Access (CDMA):

- **Concept:**

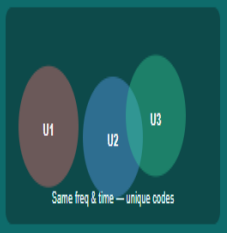
  Allows multiple users to share the same frequency band simultaneously.

  Each user is assigned a unique code to distinguish its signal from others.

- **Representation:**

  Use triangles or other shapes to represent users, with unique patterns or colors representing their codes.

  Show overlapping areas to indicate simultaneous transmission.



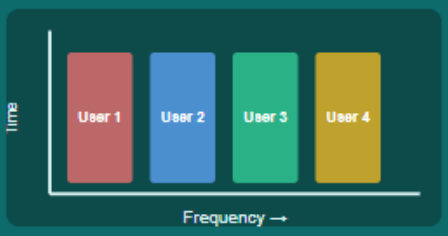
- **Key Points:**

  Users transmit simultaneously on the same frequency using different codes.

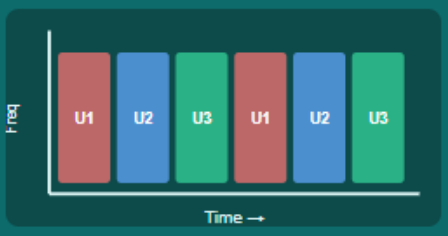
  CDMA is widely used in digital cellular networks, such as in 3G and 4G technologies.

Visual Summary

In your visual representation, you can create a comparative diagram with three sections, each highlighting the key characteristics and visual elements of FDMA, TDMA, and CDMA. Use clear labels, color coding, and arrows to illustrate differences in how these techniques allocate resources and manage multiple users on a shared medium.



FDMA



TDMA



CDMA

THE END...